



Genome-wide identification of *CAMTA* gene family members in rice (*Oryza sativa* L.) and *in silico* study on their versatility in respect to gene expression and promoter structure

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Abstract

The calmodulin-binding transcription activator (CAMTA) is a family of transcriptional factors containing a cluster of calmodulin-binding proteins that can activate gene regulation in response to stresses. The presence of this family of genes has been reported earlier, though, the comprehensive analyses of rice *CAMTA* (*OsCAMTA*) genes, their promoter regions, and the proteins were not deliberated till date. The present report revealed the existence of seven *CAMTA* genes along with their alternate transcripts in five chromosomes of rice (*Oryza sativa*) genome. Phylogenetic trees classified seven *CAMTA* genes into three clades indicating the evolutionary conservation in gene structure and their association with other plant species. The *in silico* study was carried out considering 2 kilobases (kb) promoter regions of seven *OsCAMTA* genes regarding the distribution of transcription factor binding sites (TFBs) of major and plant-specific transcription factors whereas *OsCAMTA7a* was identified with highest number of TFBs, while *OsCAMTA4* had the lowest. Comparative modelling, i.e., homology modelling, and molecular docking of the CAMTA proteins contributed the thoughtful comprehension of protein 3D structures and protein–protein interaction with probable partners. Gene ontology annotation identified the involvement of the proteins in biological processes, molecular functions, and localization in cellular components. Differential gene expression study gave an insight on functional multiplicity to showcase *OsCAMTA3b* as most upregulated stress-responsive gene. Summarization of the present findings can be interpreted that *OsCAMTA* gene duplication, variation in TFBs available in the promoters, and interactions of *OsCAMTA* proteins with their binding partners might be linked to tolerance against multiple biotic and abiotic cues.

Keywords *Oryza sativa* · *CAMTA* genes · Stress · Protein structure · Transcription factors

Introduction

Calcium (Ca²⁺) is the key regulator involved in all the vital adaptive and metabolic process of cellular responses against the environmental cues in plant system (Galon et al., 2010).

Calmodulin (CaM), CaM-like proteins (CML), and calcineurin B-like proteins (CBLs) are the three most integral classes of Ca²⁺-sensor proteins associated with the Ca²⁺-dependent regulation pathways in plants (DeFalco et al., 2010) along with Ca²⁺-dependent protein kinases (CDPKs) which act as sensor responders (Takahashi and Ito 2011). CaM is the crucial and highly researched sensor protein that can bind to the transcription factors (TFs) for regulating further metabolic, biological, and cellular processes along with transducing elicitor responses (Baum et al., 1996; Takahashi and Ito 2011). In recent years, calmodulin-binding transcription activators (CAMTAs) which are the members of the family of CaM-binding TFs or CaM-binding proteins (CBPs) are also being characterized in plant kingdom (Pant et al., 2018). The first report of CAMTA was on early ethylene-responsive gene (*NtERI*) from *Nicotiana tabacum* (tobacco) which binds to the CaM in Ca²⁺-dependent pathway to

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initiate early senescence in plant (Yang and Poovaiah, 2000). Further studies revealed the evolutionary conservation of CAMTA proteins consisting of multiple functional domains in most of the plants. CAMTA proteins possess several key functional domains like sequence-specific DNA binding domains, ‘CG-1’ comprised of bipartite nuclear localization signal available in N-terminal residue, the transcription factor immunoglobulin (TIG) domain which non-specifically attach with DNA, ‘Ankyrin repeats’ which is the key domain of protein–protein interactions, and ‘IQ’ motifs as well as ‘CaMBD’ (CaM binding domain), which respectively bind to the CaM in a Ca^{2+} -independent and dependent manner (Bähler and Rhoads, 2002; Finkler et al., 2007; Du et al., 2009). *Arabidopsis* species (*A. thaliana*, *A. lyrata*), and other plant species (*Capsella rubella*) reported to contain CAMTAs without TIG domain but the universality of the isoform is still unknown (Rahman et al., 2016). All the domains are predicted to exhibit numerous sets of activities related to regulation of gene expression.

Details about the involvement of CAMTA proteins in biological pathways are well-versed for several plant species. Six major CAMTA genes, identified from *Arabidopsis*, are associated with the stimulus response, viz., temperature extremities (Kim et al., 2013), dehydration (Pandey et al., 2013), wounding, pathogenic invasion (Nie et al., 2012), as well as ABA and ethylene signalling (Reddy et al., 2000). In a similar fashion, seven CAMTAs from *Solanum lycopersicum* (tomato) are connected with drought, cold, wounding stress, pathogenic aggression, and having key regulations in fruit development and ripening (Yang et al., 2012, 2013). Domains of CAMTAs recognize the 6 bp CG-specific binding cis-elements (G/C/A)CGCG(T/C/G) and (C/A)CGTGT present in the targeted gene promoter regions and bind there for gene expressional regulation to comprehend functional activities (Yang and Poovaiah, 2002; Du et al., 2009; Wei et al., 2017).

TFs are the backbones for initiating any transcription process after binding to the promoter region of the signalling genes for modulating growth, development, and defense mechanism in plants (Khan et al., 2018). Basic TFs, viz., AP2/ERF (APETALA 2/ethylene-responsive element binding factor), bHLH (basic helix-loop-helix), bZIP (basic leucine zipper containing domain protein), MYB (myeloblastosis viral oncogene homolog proteins), and WRKY (N-terminal WRKY domain containing TF), and additional TFs, like C2H2, CG-1, TCP (Teosinte branched1/Cinnaminate/proliferating cell factor), are intricately related to plant growth and immunity (Alves et al., 2014; Tripathi et al., 2014; Tsuda and Somssich, 2015) due to their crucial binding in the transcription factor binding sites (TFbs) available in the promoter regions of several genes.

Rice (*Oryza sativa* L.) is the chief energy source for almost 3.5 billion people globally (Najeeb et al., 2020). Rice

shares close syntenic relationship with other cereals and it is considered the model crop for grass family (Sasaki and Burr, 2000). A remarkable study in rice reported isolation of CaM-binding TF from cDNA expression library (Choi et al., 2005). Choi and colleagues identified several domains available in putative rice CaM-binding TF (OsCBT) along with the preferentially DNA-binding sequence of OsCBT (Choi et al., 2005). Loss-of-function mutation of CAMTA gene (*oscbt-1*) revealed improved resistance against bacterial leaf blight (*Xanthomonas oryzae* pv. *oryzae*) and blast fungus (*Magnaporthe grisea*), which showed the negative regulation of OsCBT to biotic stresses and confirmed it as a significant factor for mediating defense-related gene transcription (Chung et al., 2020; Koo et al., 2009). However, to the best of our knowledge, the genome-wide identification of *OsCAMTA* genes and its available transcript variants as well as detail bioinformatics study of *OsCAMTAs* including their gene promoters has not been conducted so far.

Protein complexes are often the primary factors of several cellular processes. To understand the biological machinery, we need to know the comprehensive description of the interactions of proteins, the overall three-dimensional (3D) structure, protein complexes and networks (Fuller et al., 2009; Nim et al., 2016). Based on all the information, the proteins can be modulated and prediction of the 3D structures of proteins can be done with the help of different homology modelling and molecular docking techniques (Rasheed et al., 2020). Homology modelling can give reliable structures of the binding partners and docking methods can provide the 3D model of the protein complex of interacting molecules with the help of complimentary geometrical and physico-chemical aspects (Bertoni et al., 2017; Peterson et al., 2018). Experimentally verified full-length structures of *OsCAMTA*, *OsCML*, and *OsCaM* proteins are not available in public database (pdb), and hence, it is necessary to model those proteins for the proper representation of the interaction between these proteins.

Keeping pace with the backdrop, this present in silico study has been planned to decipher the location of CAMTAs in rice genome and identify their alternate transcript generation, domain structure, evolutionary significance, availability of TFbs in their promoter region, and gene expression data in response to biotic and abiotic stresses as well as prediction of protein model and molecular docking of rice CAMTAs.

Materials and methods

Data retrieval of rice CAMTA sequences and bioinformatics analyses

CAMTA protein sequences of *O. sativa* subsp. *japonica* cv. Nipponbare were collected from both National Center for

Biotechnology Information database (<https://www.ncbi.nlm.nih.gov/>) (Jenuth, 2000) and Rice Genome Annotation Project Database (<https://rapdb.dna.affrc.go.jp/>) (Sakai et al., 2013). All available data was retrieved from NCBI database and then BLAST query analysis was executed for confirmation of *CAMTA* genes in *O. sativa* subsp. *japonica* cv. Nipponbare. The mRNA sequences along with the coding DNA sequence (CDS) which correspond to each *OsCAMTA* protein were also identified from public database. Physico-chemical characteristics of rice *CAMTA* (*OsCAMTA*) proteins such as protein size, molecular weight, isoelectric point (pI), aliphatic index, instability index, and grand average of hydropathicity index (GRAVY) for discerning hydrophobicity value of the protein were assessed by ProtParam (<http://www.expasy.org/tools/protparam.html>) tool (Gasteiger et al., 2005). Various online tools, viz., WoLF PSORT (<http://www.genscript.com/wolf-psort.html>) (Horton et al., 2007) and CELLO v 2.5 (<http://cello.life.nctu.edu.tw/>) (Yu et al., 2006), were used for prediction of potential subcellular locations of the considered rice *CAMTA* proteins. Total identified eighteen *CAMTAs* were arranged into seven groups, and for the subsequent studies of *OsCAMTAs*, only one variant (having largest protein size) has been considered from each group.

Deployment of Rice Database Oryzabase-Shigen (<https://shigen.nig.ac.jp/rice/oryzabase/>) coupled with the Chromosome Map Tool (<http://viewer.shigen.info/oryzavw/maptool/MapTool.do>) was used to construct a chromosomal map for depicting the chromosomal organization of *OsCAMTA* genes. Recognized *OsCAMTA* gene structures were analyzed using *CAMTA* CDS sequences with their corresponding gene sequences in GSDS online tool (<http://gsds.cbi.pku.edu.cn/>). The extracted sequences were analyzed to make Hidden Markov Model (HMM) profile (Finn et al., 2011) and further assessed to confirm the existence of conserved *CAMTA* domains, i.e., ‘CG-1’, ‘TIG’ domain, ‘Ankyrin repeats’ (ANK), and ‘IQ’ domain utilizing InterProScan (<http://www.ebi.ac.uk/interpro/search/sequence-search>) and Pfam (<http://pfam.xfam.org/>) database (Punta et al., 2012). CaMBD and NLS were explored by employing Calmodulin Target Database (<http://calcium.uhnres.utoronto.ca/ctdb/ctdb/home.html>) and Motif scan (https://myhits.sib.swiss/cgi-bin/motif_scan), respectively. All results were projected to construct the schematic diagram of *OsCAMTA* protein domain structure with the help of biological sequences software (<http://ibs.biocuckoo.org/>) illustrator (Liu et al., 2015).

Phylogenetic analysis of rice and Arabidopsis CAMTA proteins.

In order to carry out diversity study, only seven *OsCAMTA* sequences were used by considering the largest protein variant of each *OsCAMTA* family members. *OsCAMTAs* were aligned in cluster X program (<http://www.clustal.org/>) along with six *Arabidopsis* *CAMTAs*

(*AtCAMTAs*) for multiple sequence alignment complying default parameters (Larkin et al., 2007). Neighbor-Joining method of Molecular Evolution Genetic Analysis (MEGA X) software (<http://www.megasoftware.net/>) was operated to construct the phylogenetic tree using 1000 iterations as bootstrap values (Kumar et al., 2018).

Analysis of TFbs

Promoter region of *OsCAMTAs* was analyzed to locate the TFbs in the selected region (2.0 kb upstream region from the translation start site (TLS) of each *OsCAMTA*) through PlantPAN 2.0 software (<http://plantpan2.its.ncku.edu.tw/>) (Chow et al., 2016). Due to the presence of alternate transcript in most of the *OsCAMTAs* and variability in 5'-untranslated region (UTR) among them, for each *OsCAMTA* coding DNA sequence (CDS), the upstream 2.0 kb from TLS was designated as putative promoter region and categorized into four regions namely proximal promoter region (TLS to 500 bp upstream), first median promoter region (501 bp to 1000 bp upstream), second median region (1001 bp to 1500 bp upstream), and distal promoter region (1501 bp–2000 bp upstream). TFbs for different TFs, viz., AP2/ERF, bHLH, bZIP, C2H2, CG-1, TCP, MYB, and WRKY, were studied in considered *OsCAMTA* gene promoters. Occurrence of similar TFbs for each TF available in *OsCAMTA* promoter regions was assembled with the help of multiple promoter analysis programme (http://plantpan2.its.ncku.edu.tw/gene_group.php?# multipromoters).

Protein structure prediction

The collected protein sequences of the *OsCAMTA* genes were processed in both the platforms – iTasser (Yang and Zhang, 2015) which was developed by the Zhang lab in University of Michigan and Swiss-Model® maintained by Swiss Institute of Bioinformatics (Kiefer et al., 2009). First the protein sequences were analyzed on the iTasser online web-tool platform (<https://zhanglab.ccmb.med.umich.edu/I-TASSER/>). The top result was selected on the basis of Iden1 and Iden2 scores and also considering the coverage score (Cov score). The PDB template hits were also recorded. The top two templates were further used in Swiss-Model® (<https://swissmodel.expasy.org/>) and subsequently the top result was selected as template to build the model on the basis of Global Model Quality Estimate (GMQE) scores and identity percentage. Finally, the best template was selected and used in iTasser to construct the model for visualizing the differences and further apprehending the comparisons.

Homology modelling and docking between OsCAMTA, OsCML/OsCaM proteins

For further homology modelling, experimentally verified full-length structures of OsCAMTA, OsCML, and OsCaM proteins which are not available in pdb were also modelled using the Swiss-Model® server. The protein sequences used for generation of tertiary models were acquired from UniProt database (<https://www.uniprot.org/>). The homology modelled structures obtained from Swiss model server, were used for rigid body docking in PatchDock server (<https://bioinfo3d.cs.tau.ac.il/PatchDock/>) (Islam et al., 2021). All seven CAMTA proteins were docked (OsCAMTA1, OsCAMTA3a, OsCAMTA3b, OsCAMTA4, OsCAMTA7a, OsCAMTA7b, and OsCAMTA10) with different OsCAM and OsCML proteins (OsCam1-1, OsCam2, OsCam3, and OsCML1). The clustering root mean square deviation (RMSD) value was defaulted as 4 for every individual docking. Prediction for each individual docking was done after comparing total 20 docked models with their respective docking scores. All the docked models obtained from PatchDock server were processed for further refinement using FireDock server (<https://bioinfo3d.cs.tau.ac.il/FireDock/>) (Islam et al., 2021). The Fire Dock server utilized energy function such as atomic contact energy (ACE), van der Waals interactions, partial electrostatics, H-bond and disulfide bond, π stacking and aliphatic interactions etc. Refinement structures were ranked on the basis of energy functions and the first ranked solution of FireDock is considered to be the most stable docked model. After dock refining, prediction of Gibbs free energy (ΔG) was performed, with the help of the pyDockWEB software package (<https://life.bsc.es/pid/pydockweb>) (Jiménez-García et al., 2013). At the end of the homology modelling, the analyzed CAMTA protein structures were validated through ERRAT server (<https://servicesn.mbi.ucla.edu/ERRAT/>) (Colovos et al., 1993; Sarwar et al., 2019) and verified in PROCHECK (<https://servicesn.mbi.ucla.edu/PROCHECK/>) (Laskowski et al., 1993; Sarwar et al., 2019) to determine the scores.

Gene ontology (GO) annotation

Rice CAMTA genes were subjected to Blast2GO (newly named, OmicsBox) software (<https://www.blast2go.com/>) for analyzing gene ontology where the basic factors considered for GO annotations were molecular functions (MF), biological processes (BP), and cellular components (CC) (Conesa and Götz 2008).

Expression study of OsCAMTA genes

Expression data of *OsCAMTA* genes were retrieved from GENEVESTIGATOR (<https://genevestigator.com/>) with

Affymatrix Rice Genome Array platform (OS_AFFY_RICE-0) and mRNA-Seq Gene Level (OS_mRNASeq_RICE_GL-2) of *O. sativa* database (Das et al., 2019). The ‘Perturbations’ tool from the software was applied to know the gene expression upon different abiotic and biotic stresses utilizing filter 2.0 folds to detect the fold change in differential gene expression. The retrieved data of these fold changes in the expression of *OsCAMTAs* in response to abiotic and biotic stresses (Supplementary material 5) were employed to construct gene expression heatmap using GraphPad Prism 9 tool (<https://www.graphpad.com/scientific-software/prism/>) with red/green color scheme where the “red” color indicates upregulation and “green” color indicates downregulation of respective genes.

Results

Identification and designating unique nomenclature of CAMTA gene family members from rice

A total of 18 putative *OsCAMTA* proteins were detected and finally grouped into 7 different CAMTAs through similarity analysis and by considering single variant for each CAMTA gene (Table 1; Supplementary material 1). Chromosomes 3 and 7 hold two CAMTA genes in each of them while chromosomes 1, 4, and 10 contained 1 CAMTA gene in each (Fig. 1a). In the present study, nomenclature of different CAMTAs available in rice genome was designated considering their availability on particular chromosome to make them unique. Chromosome 1 of rice genome contained 1 CAMTA at LOC4327253 and it has been designated as *OsCAMTA1* (Table 1). A total of 3 transcript variables as well as 3 protein variants of *OsCAMTA1* were detected and designated as *OsCAMTA1.1*, *OsCAMTA1.2*, and *OsCAMTA1.3*. Although 3 transcript variables were originated from approximately same genomic position (Table 1; Supplementary material 2), *OsCAMTA1.3* had 77 nucleotide (nt) internal deletion and 551 nt deletion at 3' end compared to *OsCAMTA1.2* while *OsCAMTA1.1* transcript variable had 47 nt internal deletion compared to *OsCAMTA1.2* (Table 1; Supplementary material 2). Additionally, the amino acid (aa) length of the 3 isoforms has also exhibited variation. Isoform *OsCAMTA1.1* was found to produce a protein of 878 aa compared to 850 aa in *OsCAMTA1.2* and only 728 aa in *OsCAMTA1.3*. Chromosome 3 contained 2 *OsCAMTAs* (LOC4331903 and LOC4333007) namely *OsCAMTA3a* and *3b* having 3 alternate transcripts as well as 3 different protein variants from each of those two loci (Table 1). Out of these three *OsCAMTA3a* alternate transcripts, *OsCAMTA3a.1* and *OsCAMTA3a.2* differed in 3 nt in the exon–intron spanning region and corresponding proteins exhibited variation of 1 aa

Table 1 Characteristics of identified *OsCAMTA* genes

Gene name	CAMTA variants	Protein ID	Transcript ID	Protein size	Nucleotide size	Promoter region	Gene ID	Locus on chromosome	Strand
OsCAMTA1	OsCAMTA1.1	XP_015649554.1	XM_015794068.2	878	3199	40,397,521–40,395,521	LOC4327253	Os01g0923600	(–)
	OsCAMTA1.2	XP_015649561.1	XM_015794075.2	850	3247				
	OsCAMTA1.3	XP_015649568.1	XM_015794082.2	728	2620				
OsCAMTA3a	OsCAMTA3a.1	XP_015631248.1	XM_015775762.2	1030	3617	4,735,791–4,733,791	LOC4331903	Os03g0191000	(–)
	OsCAMTA3a.2	XP_015631249.1	XM_015775763.2	1029	3614				
	OsCAMTA3a.3	XP_015631250.1	XM_015775764.2	829	3723				
OsCAMTA3b	OsCAMTA3b.1	XP_015631089.1	XM_015775603.2	1026	3536	15,496,886–15,498,886	LOC4333007	Os03g0388500	(+)
	OsCAMTA3b.2	XP_015631090.1	XM_015775604.2	1025	3533				
	OsCAMTA3b.3	XP_015631092.1	XM_015775606.2	1023	3530				
OsCAMTA4	OsCAMTA4.1	XP_015635911.1	XM_015780425.1	1003	3630	19,106,206–19,104,206	LOC4335664	Os04g0388500	(–)
	OsCAMTA4.2	XP_025880404.1	XM_026024619.1	893	3335				
OsCAMTA7a	OsCAMTA7a.1	XP_015646431.1	XM_015790945.2	1021	7422	18,215,191–18,213,191	LOC4343265	Os07g0490200	(–)
	OsCAMTA7a.2	XP_015646432.1	XM_015790946.2	927	3247				
OsCAMTA7b	OsCAMTA7b.1	XP_015647012.1	XM_015791526.2	1041	3570	25,785,375–25,783,375	LOC4343953	Os07g0623100	(–)
	OsCAMTA7b.2	XP_015647014.1	XM_015791528.2	870	3091				
OsCAMTA10	OsCAMTA10.1	XP_015614396.1	XM_015758910.2	1024	3636	11,933,811–11,935,811	LOC4348485	Os10g0375600	(+)
	OsCAMTA10.2	XP_015614397.1	XM_015758911.2	1023	3633				
	OsCAMTA10.3	XP_015614398.1	XM_015758912.2	822	3040				

(Table 1). The mRNA of *OsCAMTA3a.3* was found to differ significantly from rest of the 2 transcripts at 5' end (Table 1; Supplementary material 2) due to the fact that some of the exons available in *OsCAMTA3a.1* and *OsCAMTA3a.2* were not available in *OsCAMTA3a.3*. Additionally, the 5' UTR region in *OsCAMTA3a.3* was different than the other 2 variants (Supplementary material 2). *OsCAMTA3b* was found to generate 3 alternate transcripts namely *OsCAMTA3b.1*, *OsCAMTA3b.2*, and *OsCAMTA3b.3*. Out of these transcripts, *OsCAMTA3b.1* is the longest transcript (3536 nt) while *OsCAMTA3b.2* is only 3 bp shorter than that (Table 1). The difference was associated with the exon–intron junction and it resulted in only 1 amino acid variation between these 2 isoforms. The transcript *OsCAMTA3b.3* has 9 nt internal deletion compared to 2 other transcripts and it was also possibly generated due to improper splicing which led to 3 aa difference between isoform *OsCAMTA3b.1* and *OsCAMTA3b.3* (Table 1). Chromosome 4 contained only 1 *CAMTA* gene (*OsCAMTA4*; LOC4335664) having 2 alternate transcripts (*OsCAMTA4.1* and *OsCAMTA4.2*). *OsCAMTA4.2* was 323 nt shorter at 5' ends compared to the *OsCAMTA4.1* transcript while the latter one had a deletion of 28 nt compared to the *OsCAMTA4.2*. Due to such extensive modifications at transcript level, *OsCAMTA4.2* produced a 110 aa truncated protein at the N-terminal end compared to the *OsCAMTA4.1* variant (Table 1). Chromosome 7 accommodated 2 *OsCAMTAs* namely *OsCAMTA7a* (LOC4343265) and *OsCAMTA7b* (LOC4343953) having 2 variants for each of them (Fig. 1a). The *OsCAMTA7a.1* and *OsCAMTA7a.2* transcripts as well as proteins were differed significantly in

their 5' end and N-terminal end, respectively. Interestingly according to NCBI database, the *OsCAMTA7a.2* transcript variable was initiated far upstream from the start site of *OsCAMTA7a.1* transcript. Between the two *OsCAMTA7b* transcript variables, *OsCAMTA7b.2* was truncated at 5' end compared to *OsCAMTA7b.1* variable (Supplementary material 2) which led to generation of truncated protein in *OsCAMTA7b.2* variant along with N-terminal sequence variation. Chromosome 10 of rice genome had a single *OsCAMTA* gene (LOC4348485), namely *OsCAMTA10* having 3 alternate forms (*OsCAMTA10.1*, *OsCAMTA10.2*, and *OsCAMTA10.3*). Out of these, *OsCAMTA10.2* transcript variant was only 3 nt shorter than the *OsCAMTA10.1* transcript and that was found to be generated from the exon–intron spanning region. Hence, the *OsCAMTA10.2* protein variant was just 1 aa shorter than the *OsCAMTA 10.1* variant as expected without any subsequent alteration in protein sequence. Strikingly, the third transcript (*OsCAMTA10.3*) was found to be truncated from 5' end region and the CDS was originated within the 7th exon of *OsCAMTA10.1*. *OsCAMTA10.3* protein isoform was 202 aa truncated from the N-terminal portion compared to *OsCAMTA10.1* isoform (Table 1). The analysis concerning whole gene structure with exon–intron organization of seven rice *CAMTA* genes is showcased in Fig. 1b.

The information regarding physicochemical characteristics, i.e., protein size, molecular weight, pI, instability index, aliphatic index, and GRAVY, are presented in Table 2 along with details of protein ID and subcellular localization. The protein size of these explored *CAMTAs* varied from 728

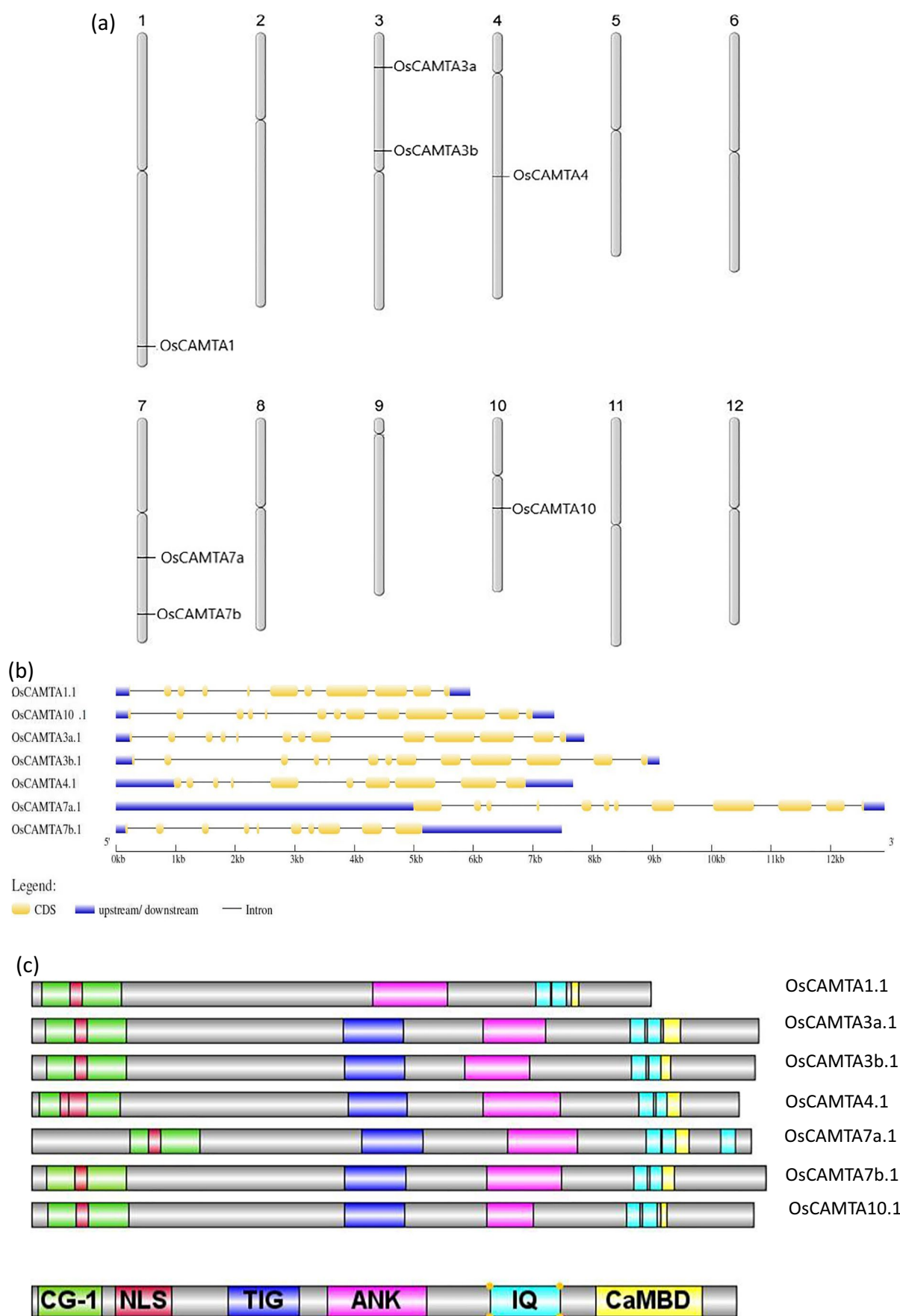


Fig. 1 **a** Chromosomal localization, **b** gene structure analysis, and **c** schematic representation of functional domains of seven different rice CAMTAs (OsCAMTA)

(OsCAMTA1.3) to 1041 (OsCAMTA7b.1) aa. The range of molecular weight and the pI differed from 81.87 kDa (OsCAMTA1.3) to 115.79 kDa (OsCAMTA7b.1) and 5.28 (OsCAMTA3a.3) to 8.43 (OsCAMTA7a.1), respectively. Correspondingly, the instability and aliphatic index ranged between 38.85 (OsCAMTA1.3) and 52.99 (OsCAMTA3a.2), and 73.50 (OsCAMTA3a.2) and 79.76 (OsCAMTA1.2). The GRAVY value was maximum for OsCAMTA7b.2 (−0.447) and minimum for OsCAMTA3a.2 (−0.566). According to the prediction of subcellular localization of OsCAMTA proteins, only OsCAMTA7b.2 was predicted to be present in chloroplast and plasma membrane along with nucleus while the rest of the OsCAMTAs were expected to be localized in nucleus (Table 2). Henceforth, only single variants from each *OsCAMTA* gene family members have been analyzed in detail.

Genome organization of *OsCAMTA* genes and analysis of protein domain and motif structure

Chromosomes possessing two *CAMTAs* were designated as ‘a’ and ‘b’ based on their location on chromosome (Fig. 1a; Table 1). Evaluation of *OsCAMTAs* revealed that all *OsCAMTA* members consisted of typical CAMTA domains and the schematic illustration of each *OsCAMTA* has been portrayed in Fig. 1c. *OsCAMTA3a.1*, *OsCAMTA3b.1*, *OsCAMTA4.1*, *OsCAMTA7a.1*, *OsCAMTA7b.1*, and *OsCAMTA10.1* were predicted to contain CAMTA-specific conserved domains such as CG-1 DNA binding domain, nuclear localization signal (NLS), TIG domain, ANK, calmodulin-binding IQ motifs, and CaMBD starting from the N-terminus to the C-terminus end (Fig. 1c). Meanwhile, *OsCAMTA1.1* contained all of the conserved domains excluding the TIG domain. All identified *OsCAMTAs* consisted of single ANK region and six of them have single NLS excluding *OsCAMTA4.1*, which have double NLS in the N-terminus. The number of IQ motifs in rice CAMTAs varied from 2 to 3. Out of the seven *OsCAMTAs*, most of them contained two IQ motifs in C-terminal residue except *OsCAMTA7a.1* possessing 3 IQ motifs. Most of the *OsCAMTAs* having CaMBD domain located adjacent to the IQ motif other than *OsCAMTA7a.1* due to presence of an extra IQ domain.

Construction of phylogenetic tree

For construction of phylogenetic tree, seven rice CAMTAs (*OsCAMTA1*, 3a, 3b, 4, 7a, 7b, and 10) were considered and aligned against six *Arabidopsis* (*AtCAMTA1*, 2,

3, 4, 5, and 6) CAMTA proteins for further comparative studies. The collective N-J tree was made to evaluate the evolutionary conservation between *Arabidopsis* and rice CAMTAs (Fig. 2a). In that tree, *OsCAMTAs* were divided into three groups (I, II, and III) based on the cluster of *AtCAMTA* and *OsCAMTA* proteins (Fig. 2a). Additionally, a separate tree of rice CAMTA also confirmed the grouping where four, two, and one CAMTA(s) were present in group I, II, and III, respectively (Fig. 2b). *AtCAMTA1*, 2, and 3 exhibited closer analogy with *OsCAMTA10*, *OsCAMTA7b*, *OsCAMTA3a*, and *OsCAMTA3b*. Furthermore, *OsCAMTA1* and *OsCAMTA4* were grouped with *AtCAMTA4* and *OsCAMTA7a* was clustered with *AtCAMTA5* and *AtCAMTA6* to represent the three distinct divisions (Fig. 2a).

Study of the corresponding TFs

The frequencies of TFbs of various TFs available in the 7 *OsCAMTA* gene promoters have been conferred in this section of the manuscript (Supplementary material 3).

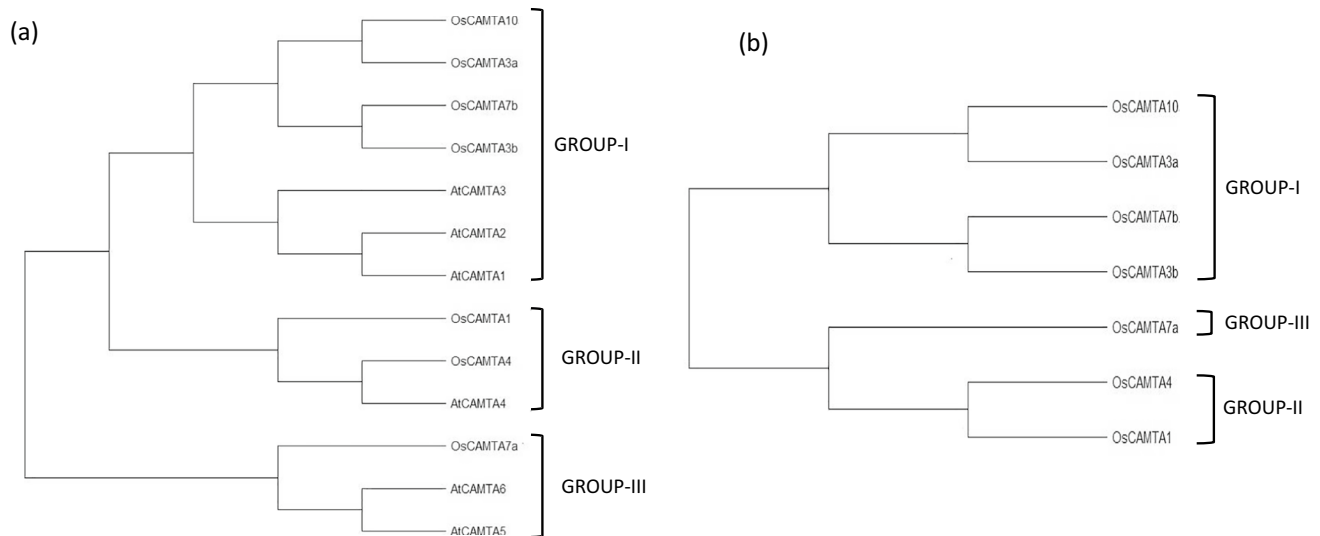
AP2/ERF A total of 606 binding sites (bs) were discovered for binding with AP2/ERF throughout the 2.0 kb analyzed promoter regions of *OsCAMTA* genes which were more or less evenly distributed across both negative and positive strands of each *OsCAMTA* gene promoters (Fig. 3a). Comparatively, meagre number (123) of AP2/ERF bs were detected from distal promoter region while a total of 176, 156, and 151 bs were detected in second median, proximal, and first median region of *OsCAMTA* gene promoters, respectively. *OsCAMTA7a* (100) and *OsCAMTA1* (96) contained maximal number of AP2/ERF bs. In the present study, AP2/ERF were exhibited to recognize distinct DNA sequences, such as ATCTT/G/C, AT(T/A)TA, ATC(A/C)A, and G/CTCTA in the promoter region of the seven *OsCAMTA* genes through in silico analyses.

bHLH The present study revealed the existence of only 53 bHLH TFbs (Fig. 3b) in the tested *OsCAMTA* gene promoter regions. Distal part of both strands consisted the highest number of bs (18), followed by second median (14), first median (12), and proximal (9) promoter region. Maximum number (12) of bHLH bs were identified in *OsCAMTA10*. The proximal promoter regions of *OsCAMTA1*, *OsCAMTA3b*, *OsCAMTA4*, and *OsCAMTA7a* were devoid of any bs while *OsCAMTA3a* and *7b* were devoid of any bHLH bs in distal promoter region (Fig. 3b). CANNTG element (E box) was detected as the most prevalent binding site among these *OsCAMTA* promoters.

bZIP A total of 68 TFbs were detected to be associated with bZIP TF across both the strands comprising of proximal,

Table 2 List of *OsCAMTA* with their physicochemical properties

Gene name	CAMTA name	Mol wt (kDa)	pI	Instability index	Aliphatic index	GRAVY	Localization
OsCAMTA1	OsCAMTA1.1	99.36	7.26	40.95	78.88	−0.533	Nucleus
	OsCAMTA1.2	96.27	7.22	39.86	79.76	−0.536	Nucleus
	OsCAMTA1.3	81.87	5.74	38.85	77.06	−0.525	Nucleus
OsCAMTA3a	OsCAMTA3a.1	115.08	5.64	52.87	73.52	−0.564	Nucleus
	OsCAMTA3a.2	115.01	5.64	52.99	73.50	−0.566	Nucleus
	OsCAMTA3a.3	91.81	5.28	46.46	74.75	−0.494	Nucleus
OsCAMTA3b	OsCAMTA3b.1	114.89	5.91	46.92	76.57	−0.512	Nucleus
	OsCAMTA3b.2	114.82	5.91	46.78	76.55	−0.514	Nucleus
	OsCAMTA3b.3	114.50	5.91	46.94	76.79	−0.513	Nucleus
OsCAMTA4	OsCAMTA4.1	112.06	5.87	48.87	75.70	−0.554	Nucleus
	OsCAMTA4.2	98.87	5.29	46.70	77.72	−0.505	Nucleus
OsCAMTA7a	OsCAMTA7a.1	113.60	8.43	49.34	74.95	−0.502	Nucleus
	OsCAMTA7a.2	103.61	7.40	45.18	78.65	−0.449	Nucleus
OsCAMTA7b	OsCAMTA7b.1	115.79	5.58	48.44	74.25	−0.529	Nucleus
	OsCAMTA7b.2	96.14	5.50	42.77	77.06	−0.447	Chloroplast, plasma membrane, nucleus
OsCAMTA10	OsCAMTA10.1	114.19	5.76	46.26	74.42	−0.528	Nucleus
	OsCAMTA10.2	114.12	5.76	46.38	74.40	−0.530	Nucleus
	OsCAMTA10.3	90.95	5.48	43.09	76.68	−0.468	Nucleus

**Fig. 2** Phylogenetic trees of *Arabidopsis thaliana* and rice CAMTA proteins. a Combined Arabidopsis and rice and b rice alone

median, and distal promoter regions of *OsCAMTA* gene promoters (Fig. 3c). Distal region was found to be the most enriched region for bZIP TFbs (25). The maximum number (17) of promoter specific sites existed in *OsCAMTA1*; however, the availability of sites was extremely low (1) in *OsCAMTA7b* which was located on the negative strand of the proximal promoter part. From the present study, it was

detected that the most distributed bs were G-boxes (CAC GTG) which acted as putative TFbs for bZIP TFs.

C2H2 Among the considered TFbs, the prevalence of C2H2 was lowest and total 27 bs were detected for C2H2 TFs among the analyzed *OsCAMATA* gene promoters (Fig. 3d), though the distribution patterns of this TFs across

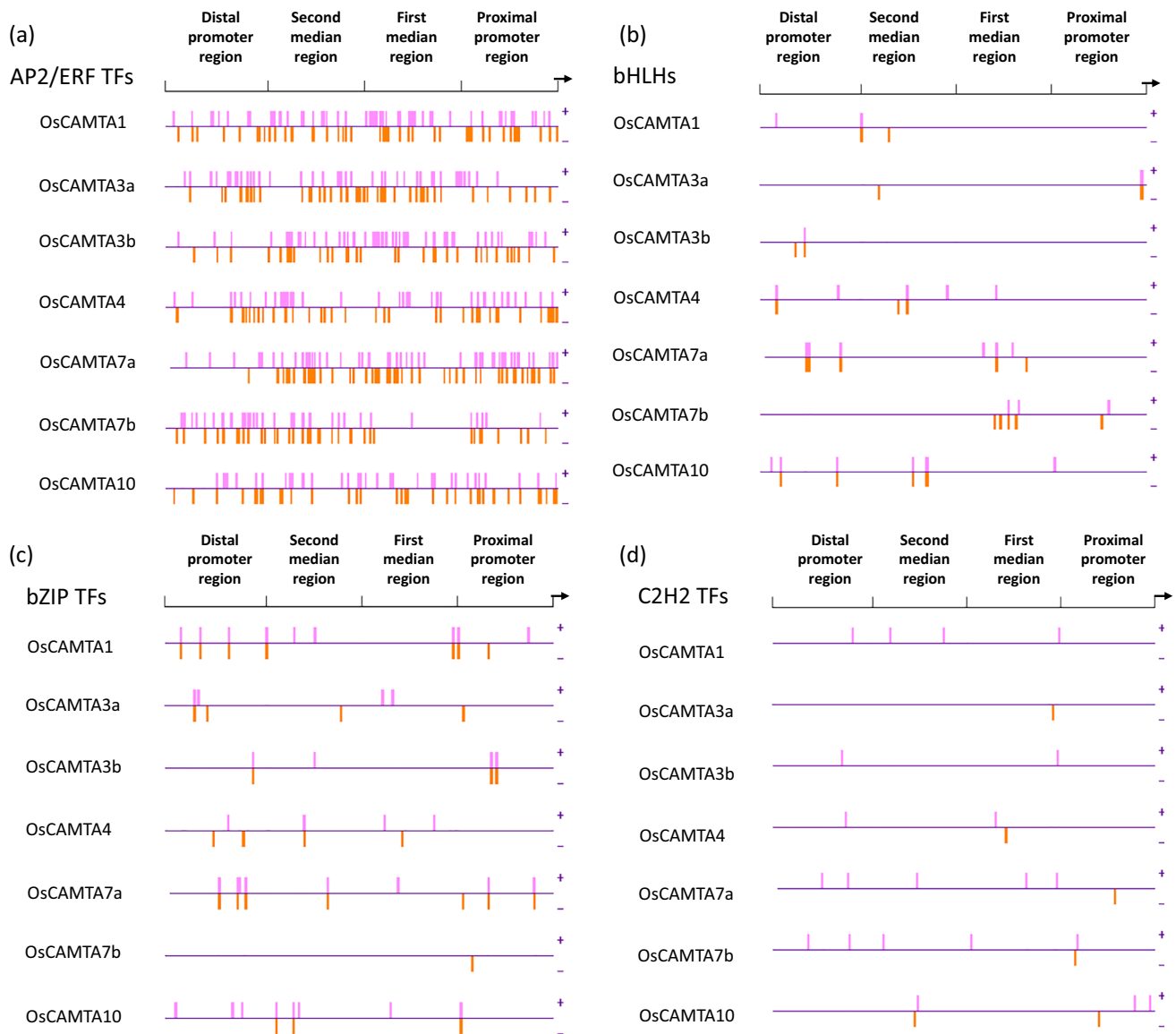


Fig. 3 Distribution of AP2/ERF, bHLH, bZIP, C2H2, CG-1, MYB, TCP, and WRKY transcription factor binding sites (TFBs) in seven *OsCAMTA* promoter regions. Putative a AP2/ERF TFbs, b bHLH TFbs, c bZIP TFbs, d C2H2, e CG-1, f MYB, g TCP, and h WRKY located on positive and negative strands of DNA are shown as pink and orange colored bars, respectively. Arrow indicates translational start site (TLS). i Relative availability of biotic and abiotic stress-related transcription factor binding sites (TFbs) in different *OsCAMTA* promoters. Numbers of TFbs in different *OsCAMTA*

AP2/ERF, bHLH, bZIP, C2H2, CG-1, MYB, TCP, and WRKY promoters are shown in respective color. The upstream 2 kilobases (kb) region from translational start site (TLS) of each *OsCAMTA* CDS was collected and analyzed as upstream regulatory region or promoter region. A total 2 kb upstream sequence was divided into four regions namely proximal promoter region (TLS to 500 bp upstream), first median promoter region (upstream 501 bp to 1000 bp), second median region (upstream 1001 bp to 1500 bp), and distal promoter region (upstream 1501 bp–2000 bp)

the illustrated regions were quiet even. Higher numbers (6 in each) of C2H2bs were observed in *OsCAMTA7a* and *OsCAMTA7b*. Only a single binding site was revealed in the promoter region of *OsCAMTA3a* for C2H2 TF. ‘NACACTN’ is the consensus DNA sequence that has been detected in our study for putative binding site of C2H2 TF.

CG1 A total of 78 CG1 TFbs were spread in both the positive and negative strands of the promoter region of *OsCAMTA* genes (Fig. 3e). Maximum 32 bs were discovered in first median region, but only 4 bs were in second median region, indicating the uneven distribution. *OsCAMTA7b* was comprised of almost 43% (34) of specified binding sites, whereas

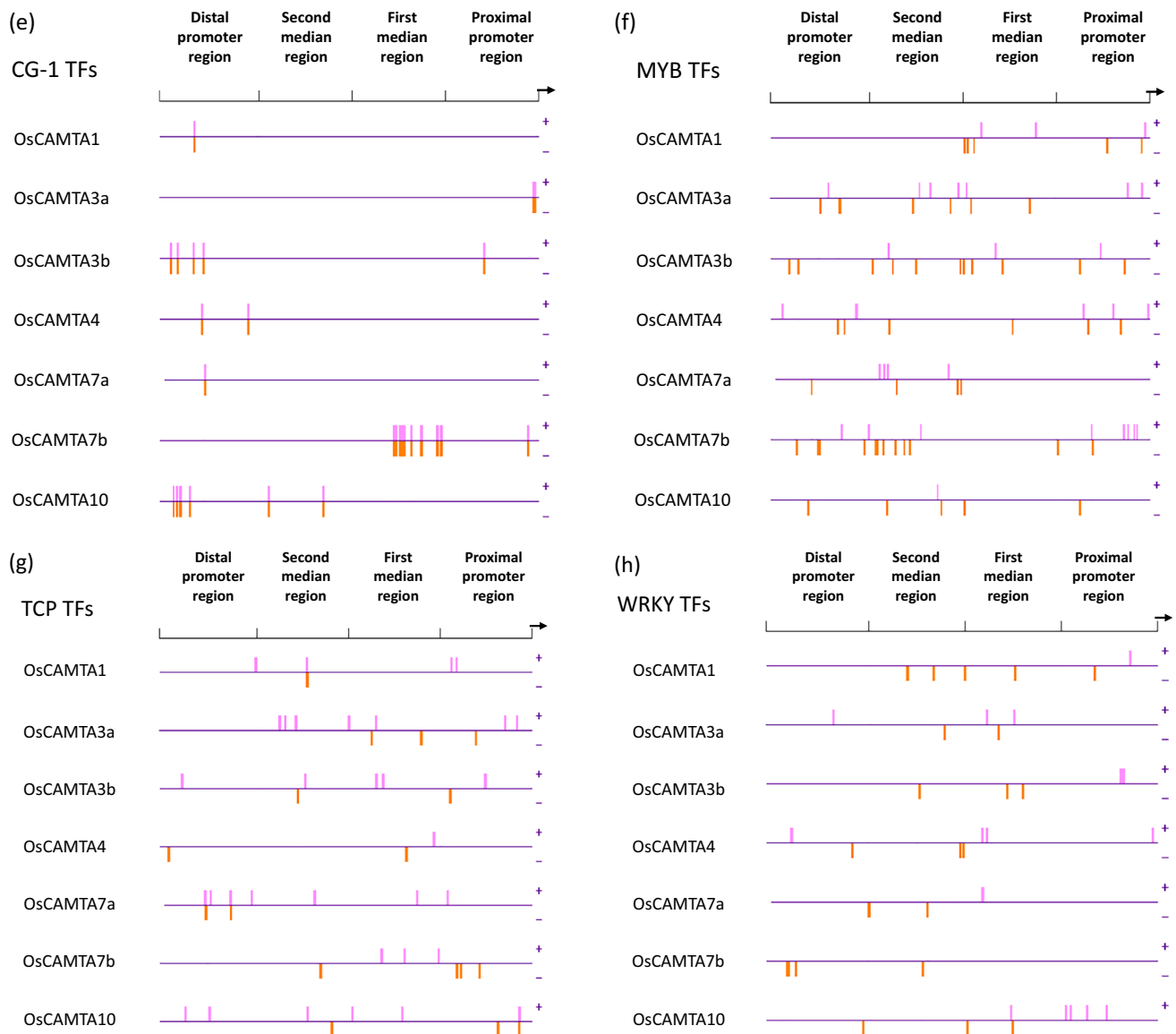


Fig. 3 (continued)

OsCAMTA1 and *OsCAMTA7a* were detected with the minimum numbers (2 in each) of CG1 bs in their promoters. Domains of 6 bp cis-elements (G/A)CGCG(C) were recognized as the putative bs for CG1 TF in the present study.

MYB In this study, total 82 MYB TFbs were identified in promoter regions of the *CAMTAs* studied in rice. Second median promoter region was detected with highest bs (29) (Fig. 3f). *OsCAMTA7b* was accommodated 20 sites, whereas the least number (6) MYBbs were found in *OsCAMTA10*. Distal region of *OsCAMTA1* was devoid of any bs while not a single MYB bs was detected in the proximal region of *OsCAMTA7a*. In our study, MYB preferred two consensus sequences, ‘AGATT’ and ‘GGATT’ mostly for binding with these seven *OsCAMTA* gene promoters.

TCP A unique plant-specific TF is Teosinte branched1/Cinnamyl/ proliferating cell factor (TCP) and its bs were identified in both the strands of proximal, median, and distal promoter regions of *OsCAMTA* genes (Fig. 3g). The total number was 50 and higher availability was noticed in the first median region (14) of promoter for *OsCAMTA* genes. The least (3) and highest (10) numbers of putative TCP bs were detected in *OsCAMTA4* and *OsCAMTA3a* gene promoters, respectively. Predominantly, abundant bs for TCPs were found to be ‘GGTCC’ in the promoter region of these *OsCAMTA* genes.

WRKY In the present study, only 38 WRKY bs were noticed in the promoter region of *OsCAMTA* members (Fig. 3h). First median region had highest number (13) of bs, trailed

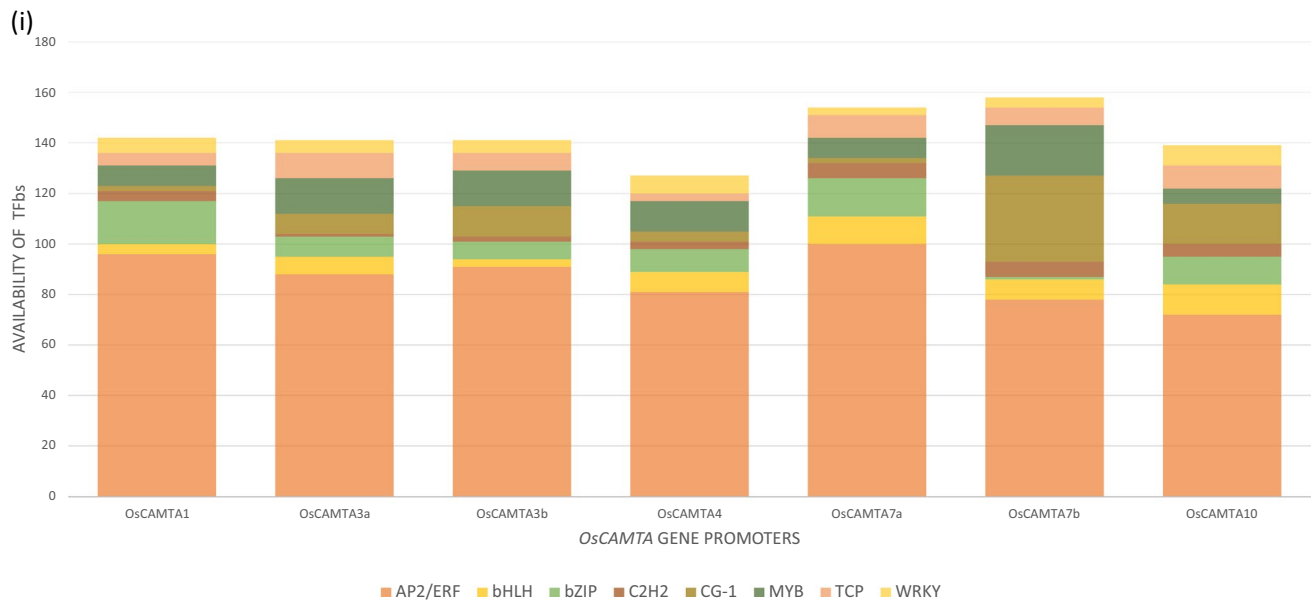


Fig. 3 (continued)

by proximal and second part of median region (9 in each) of *OsCAMTA* promoters. Relatively more bs (8) was present in *OsCAMTA10*, but *OsCAMTA7a* had least WRKY bs (3) while it did not possess any bs in distal and proximal part of the promoter. Moreover, *OsCAMTA1* and *OsCAMTA3b* lacked any site in distal region and so in *OsCAMTA3a*, *OsCAMTA7a* and *OsCAMTA7b* for proximal part. WRKY TFs were mostly bound to specific DNA binding sequence to the analyzed region having W-box containing sequence, ‘TTGACY’ (C or T replaced the letter Y) in the study.

From the current study, highest (158) and lowest (127) numbers of TFbs were observed in *OsCAMTA7b* and *OsCAMTA4* respectively, on available eight TFbs among seven *OsCAMTA* gene promoters. This analysis on locus identification of TFbs also identified *OsCAMTA3a* and

OsCAMTA3b having similar number (141) and *OsCAMTA1* possessing only one extra (142) total TFbs in the tested promoter regions. Availability of TFbs on the analyzed *OsCAMTA* genes is projected in Fig. 3i.

Analysis of protein structure modelling

All seven models from iTasser platform were designed on the basis of template PDB ID ‘5VKQA’ (Table 3) and the representative models were chosen here according to highest Cov, Iden1, and Iden2 values. The best Cov (0.97) was shown in *OsCAMTA1*, *OsCAMTA3b*, and *OsCAMTA4* (Table 3). Two models were designed in the Swiss-Model platform based on two different templates (Fig. 4). One model was constructed with selected template of best similarity percentage as well as GMQE value from Swiss-model

Table 3 Description of protein structure prediction on iTasser and Swiss-Model; Iden1: the percentage sequence identity of the templates in the threading aligned region with the query sequence; Iden2: the percentage sequence identity of the whole template chains

CAMTA Name	iTasser			Swiss-Model (selected template)			Swiss-Model (iTasser template)		
	Template	Iden1, Iden 2	Cov	Template	GMQE	Similarity %	Template	GMQE	Similarity %
OsCAMTA1	5vkqA	0.10, 0.19	0.97	1n0r.1.A	0.04	41.30	5vkq.1.A	0.04	32.98
OsCAMAT3a	5vkqA	0.12, 0.24	0.94	1qym.1	0.05	21.79	5vkq.1.A	0.04	24.41
OsCAMTA3b	5vkqA	0.13, 0.22	0.97	5hry.1.A	0.04	31.97	5vkq.1.A	0.04	24.80
OsCAMTA4	5vkqA	0.10, 0.19	0.97	5aq7.1.A	0.04	34.74	5vkq.1.A	0.03	22.77
OsCAMTA7a	5vkqA	0.13,0.23	0.94	4hll.1.A	0.04	27.34	5vkq.1.A	0.03	24.06
OsCAMTA7b	5vkqA	0.13,0.23	0.96	5lw2.1.A	0.06	26.67	5vkq.1.A	0.03	20.91
OsCAMTA10	5vkqA	0.12,0.25	0.95	6s9s.1.A	0.04	29.37	5vkq.1.A	0.03	23.08

with query sequence; Cov: the coverage of the threading alignment and is equal to the number of aligned residues divided by the length of query protein; GMQE: a quality estimate which combines properties from the target-template alignment and the template structure

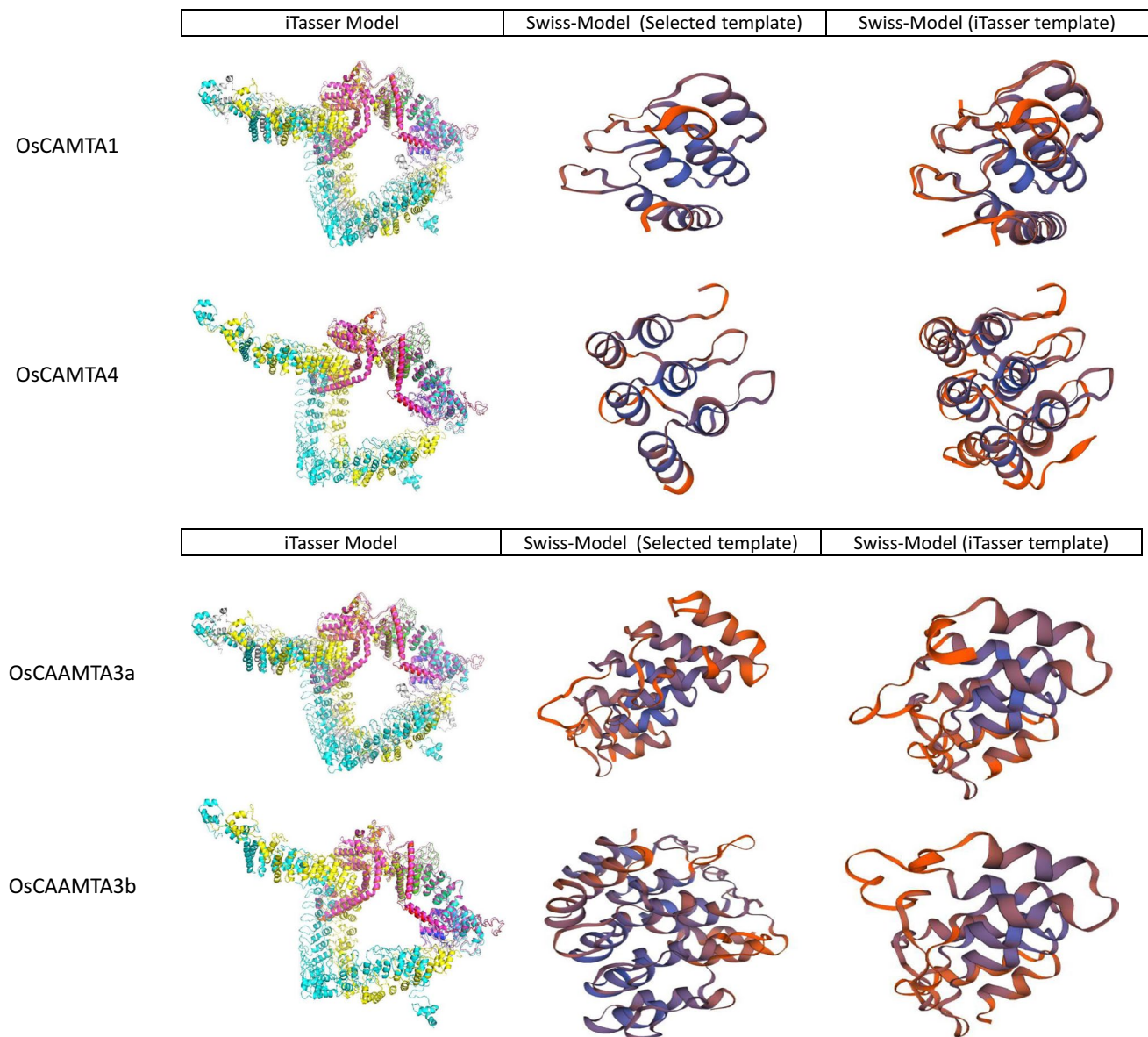


Fig. 4 Predicted protein structure of OsCAAMTA1.1, OsCAAMTA3a.1, OsCAMTA3b.1, OsCAMTA4.1, OsCAMTA7a.1, OsCAMTA7b.1, and OsCAMTA10a.1

itself and another model with i-Tasser template. In Swiss model, template with maximum similarity percentage of 41.30 was found in OsCAMTA1 and only 21.79% was found in OsCAMTA3a with the query protein sequences (Table 3).

Docking of OsCAMTA with OsCML/OsCaM proteins

For docking study, OsCaM1-1, OsCaM2, OsCam3, and OsCML proteins were considered with the tested OsCAMTAs. The top five docked models were the most stable structures, predicted by FireDock (Fig. 5). The first model was docking between OsCAMTA3b and OsCaM1-1, where docking score was highest (15,008) (Table 4).

Subsequent docked models were of OsCAMTA7a and OsCML1, OsCAMTA3b and OsCaM2, and OsCAMTA3b and OsCML1. The least docking score (14,282) was measured between OsCAMTA7a and OsCaM3. The binding residues of OsCAMTAs and OsCaM/OsCMLs are enlisted in Table 4. Docking refinement with pyDockWEB server on the basis of ΔG value was differed from FireDock analysis depicting that OsCAMTA3a and OsCML1 had the largest ΔG of -56.977 kJ/mol, while lowest ΔG of -26.787 kJ/mol were revealed by OsCAMTA7a and OsCaM3 (Supplementary material 4).

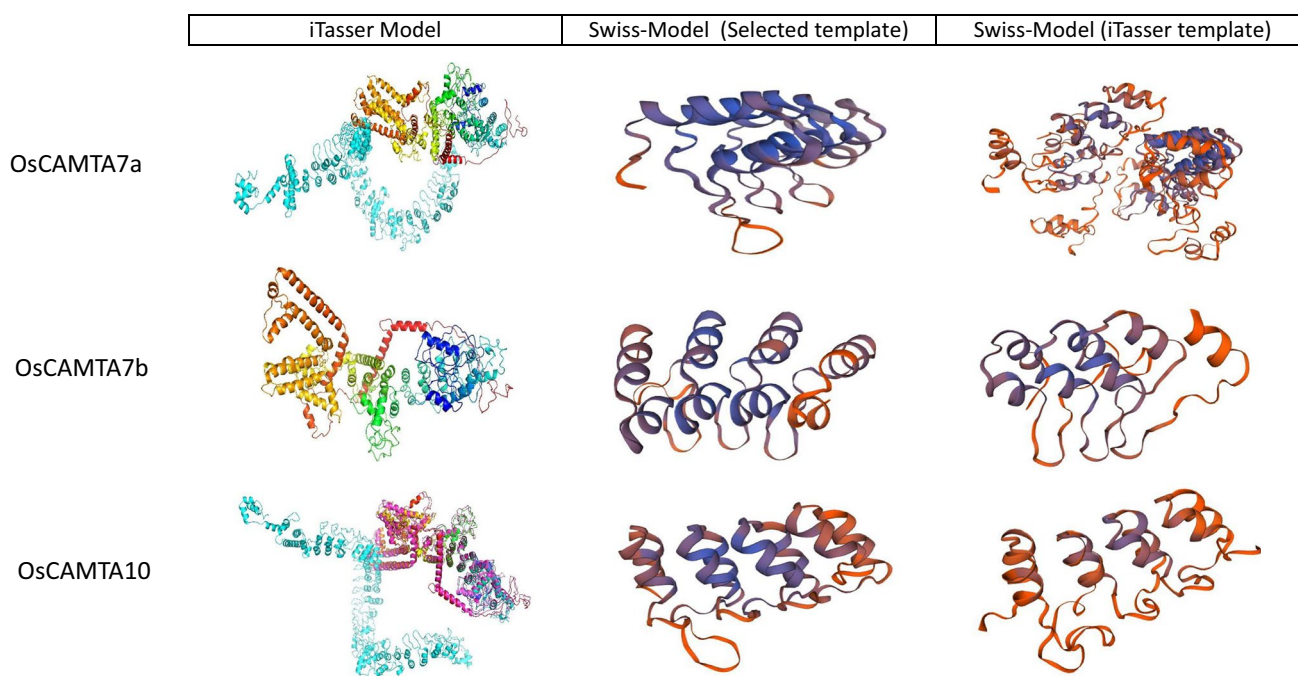


Fig. 4 (continued)

GO annotation of rice CAMTA proteins

Annotation of various function of the studied OsCAMTA proteins from rice could be categorized into 3 functions, viz., molecular, biological processes and cellular components. It was depicted that biological processes of the *OsCAMTA* genes were mainly associated with regulation of biological, metabolic, and cellular regulations, and also linked with the response to biotic and abiotic stimulus. Predominantly, molecular function was sequence-specific DNA binding and the targeted structural cellular component was nuclear (Fig. 6). The only exception for cellular component analysis was the detection of OsCAMTA7b protein as an integral component of membrane. Besides, OsCAMTA10 exclusively exhibited DNA-binding TF activity and calmodulin-binding as molecular functions analysis and cellular response to cold as biological processes based on this GO database (Table 5). Rice CAMTA annotations were primed to positive regulation of transcription by RNA polymerase II (GO:0,045,944) and response to cold stimulus (GO:0,070,417) for BP; sequence-specific DNA binding (GO:0,043,565), DNA-binding transcription factor activity (GO:0,003,700), and calmodulin-binding (GO:0,005,516) for MF; and nucleus (GO:0,005,634) and integral component of membrane (GO:0,016,021) for CC (Table 5) (Conesa and Götz 2008).

OsCAMTA gene expression under different abiotic and biotic stress conditions

OsCAMTA gene expression study under abiotic and biotic stress conditions employing *O. sativa* microarray database in GENEVESTIGATOR showed involvement with stresses. Expression of *OsCAMTA3b* gene displayed most differential expression in response to several abiotic stresses like anoxia, dehydration, drought, heat, salt, and submergence stress (Fig. 7). *OsCAMTA4* and *OsCAMTA7a* indicated upregulated expression under salt stress. Additionally, *OsCAMTA1* signified about more than 2.0 folds expression upregulation in response to drought stress (Supplementary material 5). Upon exposure to anoxia, most of the genes including *OsCAMTA3a*, *3b*, *7b*, and *10* showed downregulation (Fig. 7). Heat stress also depicted downregulation in *OsCAMTA1*, *OsCAMTA3a*, and *OsCAMTA4* gene expression. For biotic stresses, except *OsCAMTA1* other six CAMTA genes from rice revealed upregulation upon *Burkholderia glumae* infection where *OsCAMTA3b* gene was upregulated the most. On the contrary, *OsCAMTA3a* and *OsCAMTA3b* were strikingly and moderately downregulated against *Xanthomonas oryzae* infestation. Due to scarcity of available transcriptomic data on *OsCAMTA* gene expression studies, identification of similar gene family member associated with a particular stress was found to be difficult and further study is needed in this regard.

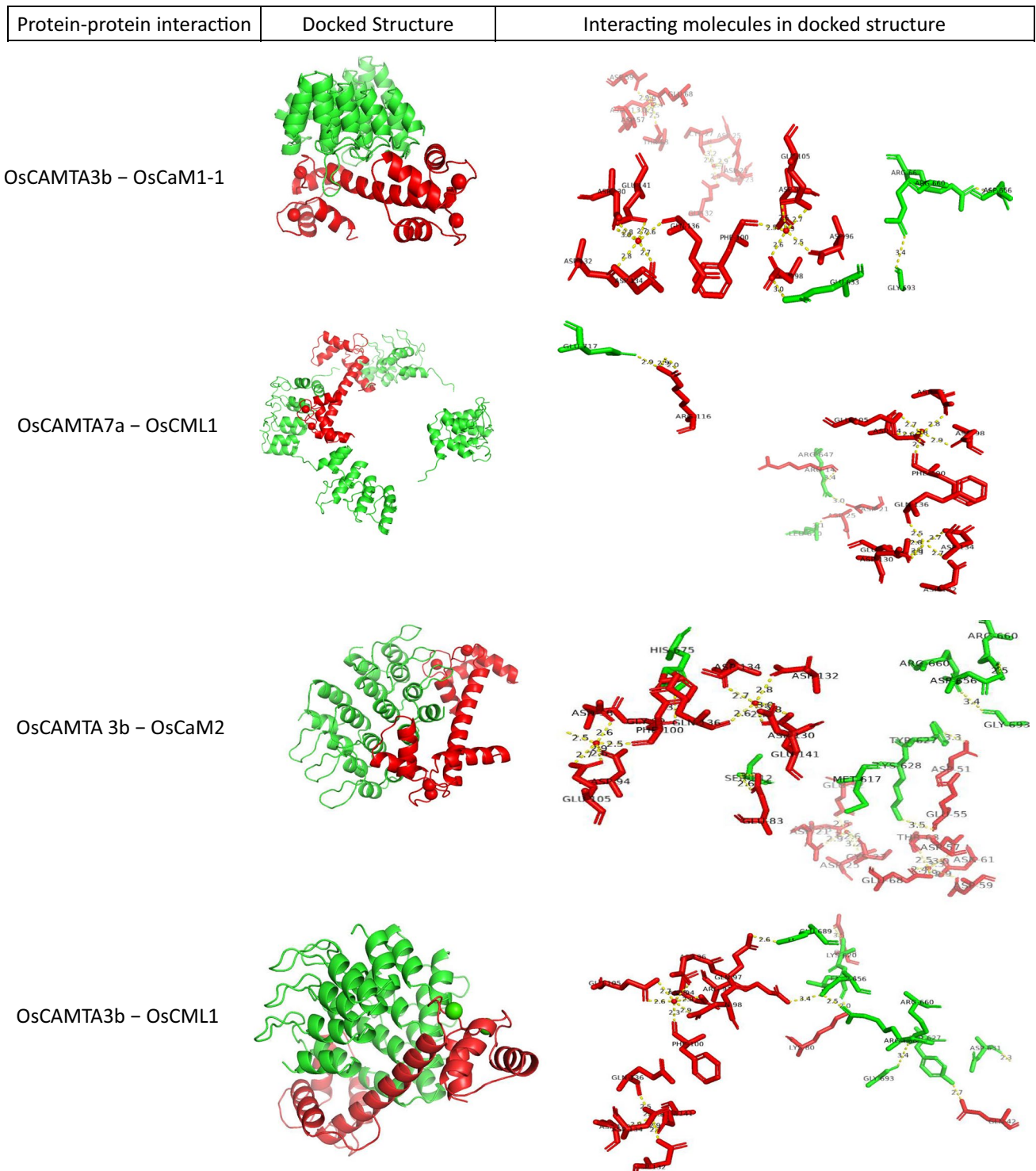


Fig. 5 Molecular docking between OsCAMTA and OsCaM/OsCML proteins. Here OsCAMTAs are represented as green color and OsCaM/OsCMLs are represented as red in color

Discussions

CaM is Ca^{2+} -binding regulator for modulating a number of target proteins while CAMTA TFs activate the gene

regulation of several Ca^{2+} /CaM signalling pathway components in response to exogenous hormonal, environmental, and stressed conditions encountered by plants (Boonburapong and Buaboocha 2007; Galon et al., 2010).

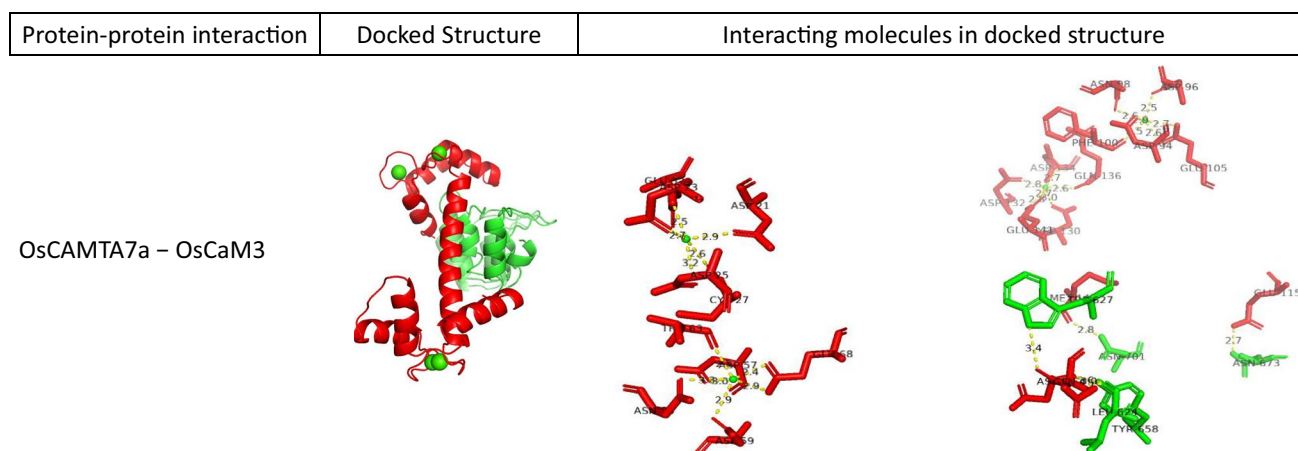


Fig. 5 (continued)

Table 4 Enlistment of interacting residues of top 5 docked structures among *OsCAMTAs* and *OsCam/OsCml* proteins

Protein–protein docking	Docking score	Interacting residues of Cam/Cml	Interacting residues of CAMTA
OsCAMTA3b–OsCaM1-1	15,008	D21,D23,D25,C27,E31,D56,D57,D59, N61,T63, E68,D93, D95, N97,F95,E104, D130,D131 D133, Q135,E140	chain A -R659,G692, chain B E632, D655, R659
OsCAMTA7a–OsCML1	14,738	R14, D21, D25, F93, D94, D96, N97, F100, E104, R115, E129, D131, D133, Q135, E140	R646, L669, E716, Y760
OsCAMTA3b–OsCaM2	14,510	R13, D20, D24, D93,95, F99, E104,D129,131,133, Q135, E140	chain B E716, Y760 Chain D – R646, L669, Q671
OsCAMTA3b–OsCML1	14,292	D21,23, D25, D27,E32, D51, E55,D57, D59, N61, T62,E67 E83, D94, N98, G99, F100, E105, D129, D131, D133, Q135 E141	chain A -R659, G692, Chain B- K619, Y626, Y629, D630, D659, E688,
OsCAMTA7a–OsCaM3	14,282	D3, Q4, D21, D23, D25, C27, E32, D57, D59, N60, T62, E68, D93, D95, N97, F99, E104, E114, D129, D131, D133, D135, E140, M145	L623, W626, Y657, N672, N700

CAMTA TFs are vital players in the Ca^{2+} /CaM transduction signal pathway (Yang et al., 2015). In earlier study in wheat, 15 *CAMTA* genes were identified and their detail analyses including location on chromosome along with other physicochemical properties and expression profiling were characterized elaborately which confirmed their role towards maneuvering biotic and abiotic stresses (Yang et al., 2020). An overview of 29 *CAMTA* genes in 4 *Nicotiana* species has been presented which also affirmed their roles in regulating biological process as well as mitigating stresses in plants (Kakar et al., 2018). Furthermore, in model legume Medicago, *MtCAMTAs* genes were identified to be associated with regulation of nodule organogenesis (Yang et al., 2015). In another comprehensive study, 22 *CAMTA* gene family was identified in 3 *Gossypium* species and depicted their role in cotton fiber development (Pant et al., 2018). Although, very meagre number of *CAMTA* members have been studied in rice, the current research identified the genome-wide *CAMTA* gene families available in rice along with their unique nomenclature. It was detected that chromosome

1 occupied *CAMTA4* isoform X1 (XP_015649554) and another protein of similar nomenclature “*CAMTA4* isoform X1” (XP_015635911) was also located in chromosome 4. Furthermore, chromosome 3 accommodated both *CAMTA1* (XP_015631248) and *CAMTA3* (XP_015631089) genes along with transcript variables available with them. In addition to that, 2 isoforms of *OsCAMTA1* (XP_015631248 and XP_015631249) differed only for single aa and the transcript was found to vary in exon–intron junction. Hence, pressing need is felt towards proper nomenclature to distinguish between different *CAMTAs* available on particular chromosome coupled with their isoforms and the nomenclature used in the current study (Table 1) can distinguish them uniquely without any perplexity.

It has been demonstrated that CG-1 comprised of NLS, ANK, IPT/TIG, and IQ in *OsCAMTA* gene family are basic conserved domains of *CAMTA* family members discovered in most of the plant species (Yang et al., 2015; Kakar et al., 2018). Sequence analysis of rice *CAMTAs* depicted that only *OsCAMTA1.1* does not have any TIG domain (Fig. 1c),

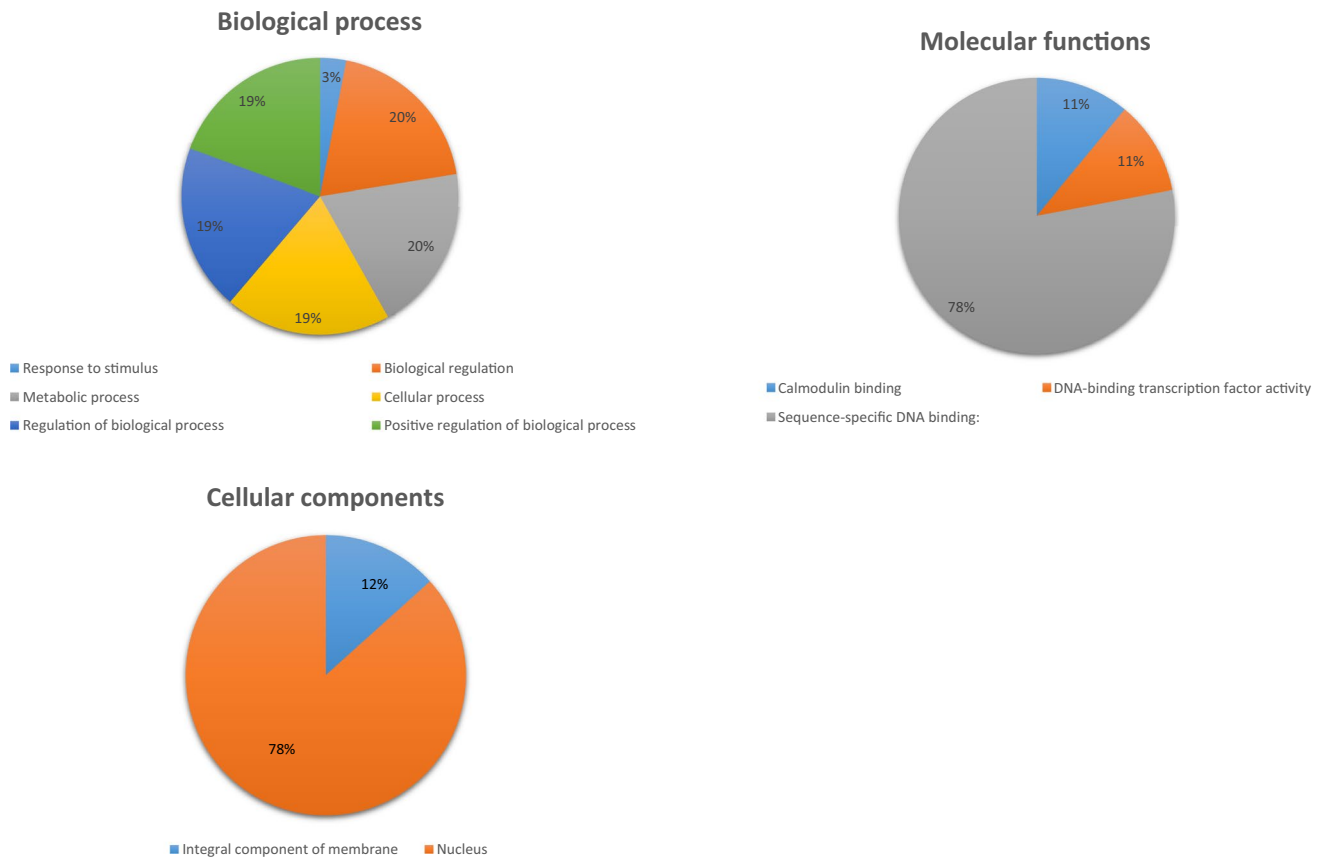


Fig. 6 GO functional annotation for the rice CAMTA genes. Cellular components, molecular functions, and biological processes are detected for OsCAMTAs. The different colors represent the proportions of various GO functions

but rest of the OsCAMTAs have all the conserved domains with nuclear localization property. The rice CAMTAs without CG-1 domain were also devoid of bipartite NLS (composed of basic amino acids arginine and lysine) in the N-terminus. It is predicted that this NLS region might constitute a signal that directs *OsCAMTAs* to the nucleus (Pant et al., 2018). Moreover, TIG domain is not universal in all CAMTAs, because evolution of non-TIG type CAMTAs is found to be the result of various mutations in that domain (Rahman et al., 2016), so the isoform might not work functionally and further study is needed in that regard.

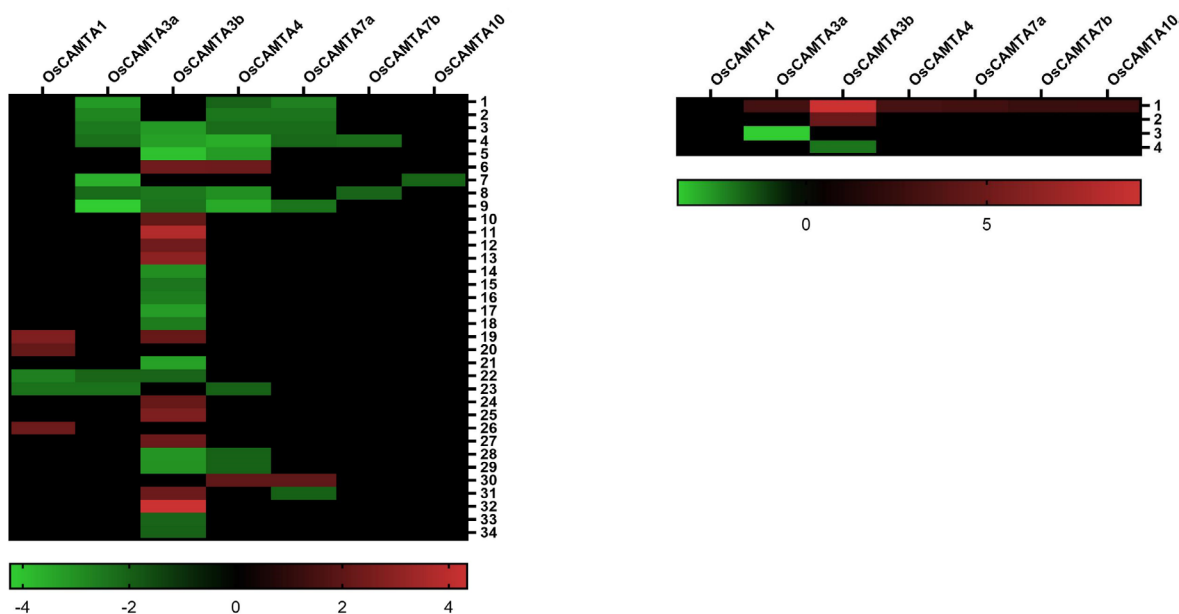
Creation of the phylogenetic tree containing *Arabidopsis* and rice orthologs indicated similar functionalities of rice CAMTAs with CAMTAs of other plant species. All of these *Arabidopsis* genes are positive regulator of cold tolerance while *AtCAMTA3* acts as negative regulator of salt tolerance and plant defense against pathogen (Doherty et al., 2009; Kim et al., 2013; Prasad et al., 2016; Qiu et al., 2012). *AtCAMTA3* was found to negatively regulate the resistance against *Xanthomonas oryzae* pv. *oryzae* (Rahman et al., 2016). In a similar fashion, *OsCAMTA3a* and *3b* expression were found to be reduced upon exposure to *X. oryzae* (Fig. 7). It is to be recalled here that *AtCAMTA3*

were grouped under same clade along with *OsCAMTA3a* and *3b* (Fig. 2). Additionally, *OsCAMTA7a* were in same group as *AtCAMTA5* and *AtCAMTA6*. *AtCAMTA6* gene also revealed significant role in regulating salt stress (Shkolnik et al., 2019) while the only characterized *OsCAMTA* (*OsCBT* which is equivalent to *OsCAMTA7a*) depicts negative regulation to biotic stresses (Koo et al., 2009) but strong involvement upon salt stress exposure (Fig. 7). Further study is needed to decipher the structural similarity of *OsCAMTAs* with their functional significance, if any.

Association with the ABA-dependent and ABA-independent pathway is the basis for classification of TF families (AP2/ERF, bHLH, bZIP, MYB, WRKY and some plant-specific TFs, viz., C2H2, CG-1 and TCP) in plant species including rice (Lindemose et al., 2013). AP2/ERF TF is related to multiple defense pathways in plants and that can be a reason for prevalence of exceptionally higher number of AP2/ERF bs in the promoter regions among the *OsCAMTA* members (Fig. 3i; Supplementary material 3). CRT elements and GCC box are the most common binding elements for AP2/ERF to tackle abiotic and biotic effectors (Singh et al., 2002). Similarly, previous experimental findings also validated the binding of bHLH TF with the E-Box sequences of

Table 5 Gene Ontology (GO) terms annotation of *OsCAMTA* genes

Protein names	Biological process IDs	Biological process names	Molecular function IDs	Molecular function names	Cellular component IDs	Cellular component names
OsCAMTA1	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565	F: sequence-specific DNA binding	C: GO: 0,005,634	C: nucleus
OsCAMTA3a	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565	F: sequence-specific DNA binding	C: GO: 0,005,634	C: nucleus
OsCAMTA3b	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565	F: sequence-specific DNA binding	C: GO: 0,005,634	C: nucleus
OsCAMTA4	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565	F: sequence-specific DNA binding	C: GO: 0,005,634	C: nucleus
OsCAMTA7a	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565	F: sequence-specific DNA binding	C: GO: 0,005,634	C: nucleus
OsCAMTA7b	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565	F: sequence-specific DNA binding	C: GO: 0,005,634 C: GO: 0,016,021	C: nucleus C: integral component of membrane
OsCAMTA10	P: GO: 0,045,944	P: positive regulation of transcription by RNA polymerase II	F: GO: 0,043,565 F: GO: 0,003,700	F: sequence-specific DNA binding F: DNA-binding transcription factor activity	C: GO: 0,005,634	C: nucleus
	P: GO: 0,070,417	P: cellular response to cold	F: GO: 0,005,516	F: calmodulin binding		

**Fig. 7** Heatmap analysis of *OsCAMTA* genes under abiotic and biotic stresses. Color bar represents fold change in gene expression where green color represents lowest level of expression and red represents highest level of expression

CANNTG (Chang et al., 2015), which was also detected in this study. Earlier, it was presented that G-box, C-box (GAC GTC), A-box (TACGTA), and some palindromic sequences are suitable TFbs for bZIP (Jakoby et al., 2002) and G-boxes (CACGTG) are indicated to be prevalent in *OsCAMTA*s gene promoters also. MYB recognition elements (MREs) are cis-acting regulatory elements, recognized as A(A/C) C(A/T)A(A/C)C like consensus sequences based on the earlier reports (Rushton and Somssich, 1998). ‘CAACTG’, ‘CNGTTR’, ‘TAACTG’, ‘GGATA’, ‘YAACKG’, ‘AGATT’, ‘GGATT’, ‘CGATT’, and ‘TGATT’ are other verified sequences for MYB to bind for specific gene expression (Hosoda et al., 2002; Das et al., 2019). MYB specified to bind to the regions of ‘AGATT’ and ‘GGATT’ of these *OsCAMTA* promoters. The minimum prerequisite of WRKY TFs for specific DNA binding to the targeted gene is ‘TTGACY’ (C or T replaces the letter Y), which are prevalent in *OsCAMTA* promoter regions. Some earlier researches demonstrated the presence of ‘TTGACG’ or ‘TTGACA’ for binding of WRKY in its promoter (Kankainen and Holm, 2004; Qiu et al., 2007). These stress-responsive TF families, i.e., AP2/ERF, bHLH, bZIP MYB, and WRKY, are the epitome of involvement and improvement in biotic and abiotic stress tolerance in plants (Yoon et al., 2020). Plenty of published reports on these proved the impact of these TFs in plant development and defense mechanism, such as AP2/ERF in drought, salt, flooding, and temperature extremities (Kasuga et al., 1999); bHLH in seed germination, anthocyanin biosynthesis, flowering, drought, cold, salinity, iron uptake, and utilization (Gao et al., 2019; Ogo et al., 2006; Dombrecht et al., 2007; Wang et al., 2015); bZIP in ABA-dependent stress metabolism (Fujita et al., 2013); MYB in extreme temperatures, drought, salt, phosphorous starvation, metal, sugar, light, and wounding stress (Li et al., 2015) ABA and SA metabolism (Ambawat et al., 2013); and WRKY in plant and root growth, germination, senescence, seed development, and biosynthesis of alkaloids (Nan and Gao, 2019) along with hot as well as cold temperatures, dehydration and salt stress (Chen et al., 2012). Moreover, AP2/ERF (Lu et al., 2011; Liu et al., 2012), bHLH (Ogo et al., 2006; Dombrecht et al., 2007; Wang et al., 2015), bZIP (Singh et al., 2002), MYB (Ambawat et al., 2013; Kranz et al., 1998; Mengiste et al., 2003; Vannini et al., 2007), and WRKY (Liu et al., 2005; Eulgem et al., 2000) also play an important role in biotic stress management against bacteria, virus, fungi, herbivores, and other harmful pathogens to plants. As an important TF for plant development and defense, member of C2H2 zinc finger family binds to the various elements, e.g., A(G/C) T repeat sequences, A[AG/CT]CNAC, TACAAT motifs. ‘NACACTN’ is the consensus DNA sequence that has been spotted for *CAMTA*s gene promoters in rice. C2H2 proteins act as leading factors in response to high temperature, salt, osmotic, cold, drought, and oxidative stimuli (Han et al.,

2020). Report confirmed that tissue-specific gene expression was governed by these TFs for biotic stress management through inducing SA and Jasmonic Acid (JA) signalling pathway (Yin et al., 2020). Generally, in earlier studies, the potential bs for TCP were ‘GGNCCCAC’, ‘GTGGNCCC’, ‘GTGGNCCCNN’, and ‘GTGGGNCCCNN’ consensus sequences (Viola et al., 2012) and in the present finding the bs were perceived to ‘GGTCC’ in the promoter region of the *OsCAMTA* genes. This crucial TF also participates in effector-triggered immunity (ETI) for saving the plants from insects and other biotic stimulus (Ling et al., 2020). Moreover, TCP has some ability to regulate responses against salinity and dehydration (Ling et al., 2020). Earlier, systematic evaluation on *OsCAMTA* gene promoters was not done regarding abiotic and biotic stress-related TFbs. To know more about the biological application of rice *CAMTA*s, further comprehensive research is required on the promoters with high and low TFbs.

Swiss model is usually widely used online platform for doing protein homology modelling. Due to lack of whelming homology, iTasser (Yang & Zhang, 2015) was also chosen along with Swiss-Model (Kiefer et al., 2009). It was reported that iTasser was better in accuracy in these cases as it starts ‘ab initio’ modelling where there is lack of homology. The comparisons of the results (Table 3) obtained from both the platforms will result in a better understanding about these homology modelling techniques as it will be more reliable to generate and compare between two widely used platforms (Fig. 5).

After getting reliable structures of the binding patterns of protein from homology modelling, the molecular docking techniques can offer the 3D model of the protein complex with the interacting molecules based on complimentary geometrical and physicochemical characteristics (Bertoni et al., 2017; Peterson et al., 2018). *OsCaM* and *OsCML* are a multi-functional protein, which can sense the increased Ca^{2+} concentration and transduces signal in the development and defense mechanism in abiotic and biotic stresses, accordingly (Boonburapong and Buaboocha 2007; Chinpongpanich et al., 2012; Shi and Du 2020), and *CaMs/CMLs* are responsible for binding and subsequent regulation of downstream target proteins such as *CAMTA*s or *CBTs*, which are required in plant responses through *CaM/CML* binding network (Zeng et al., 2015; Patra et al., 2021). Homology modelling and molecular docking techniques would provide valuable insights on these protein–protein interaction and application on the plant signalling mechanisms. Till now, total of 32 *CML* proteins and 5 *CaM* proteins have been identified in rice (Boonburapong and Buaboocha 2007). Phylogenetic study says *OsCaM1-1*, *OsCaM1-2*, *OsCaM1-3*, *OsCaM2*, and *OsCaM3* share > 97% identity while *OsCaM1-1*, *OsCaM1-2*, and *OsCaM1-3* encode same protein (Boonburapong and

Buaboocha 2007). So, OsCaM1-2 and OsCaM1-3 could not be considered for docking study. OsCML1, OsCML2, OsCML3, OsCML4, and OsCML5 share > 60% identity. For docking study, OsCaM1-1, OsCaM2, OsCaM3, and other OsCML proteins were contemplated with the tested seven OsCAMTA proteins. *OsCam1-1* gene can confer tolerance under salt stress and heat with the help of biphasic Ca^{2+} signal (Saeng-ngam et al., 2012; Yuenyong et al., 2018; Kaewneramit et al., 2019; Phean-O-Pas et al., 2005; Wu et al., 2012). Furthermore, investigation in GENEVESTIGATOR indicated downregulation of *OsCAMTA3b* in presence of salt stress and differential regulation under high temperature (Fig. 7). Molecular docking on these two proteins (OsCaM1-1 and OsCAMTA3b) showed the highest score (Table 4) which represented the complementation function. Similarly, increased amount of *OsCaM3* was expressed in 1 to salt, mannitol, and wounding (Phean-O-Pas et al., 2005). In the present study, OsCAMTA7a proteins showed better docking score with OsCaM3 (Table 4; Fig. 5) and portrayed upregulation under salt stress (Fig. 7). Another study suggested *OsCML1* had induced expression after receiving salinity stimuli (Chinpongpanich et al., 2012) which is corresponding to the expression of *OsCAMTA7a* as represented in the study (Fig. 7, Supplementary material 4).

Gene duplication and variation with respect to their regulation is the subsequent event of evolution and present study depicted homology in gene expression of *OsCAMTAs*. Interpretation of the present study with other *CAMTA* genes (Qiu et al., 2012; Shkolnik et al., 2019; Kim et al., 2013) depicted that *OsCAMTA* genes as well as their promoters might be involved in resistance mechanism under both abiotic and biotic stresses in plants and probable interactions of *OsCAMTAs* with *OsCAM/CML* could be the major determining factor in stress recognition as well as subsequent downstream signalling (Patra et al. 2021). Further study is needed to unravel the *CAMTA*-specific signalling cascade and identify their explicit binding partners to combat plant stressors.

Conclusion

The detailed in silico analysis on *CAMTA* genes of rice (*O. sativa*) gave some insights on the structure and function of the genes. In the present study, total of seven genes of *OsCAMTA* family with their possible transcripts (complete 18 genes) were detected and characterized. The phylogenetic evolution and similarities with other plant species in association with the expression study from GENEVESTIGATOR (<https://genevestigator.com/>) with Affymatrix Rice Genome Array platform (OS_AFFY_RICE-0) and mRNA-Seq Gene Level (OS_mRNASeq_RICE_GL-2) of *O. sativa* database envisaged towards the functional similarities

and projected the conservation of the genes. Comparative analysis of predicted protein structures was demonstrated and molecular docking was done for the better understanding of the protein–protein interaction of *CAMTAs* with *CaMs/CMLs* of rice. All the considered *OsCAMTAs* gene promoters possess difference in terms of the presence of TFbs where *OsCAMTA7a* and *OsCAMTA4* revealed highest and lowest TFbs, respectively. These genes will be significant for future studies in genetic engineering for signalling and stress management. Gene ontology on *CAMTA* genes validated the urgency of further characterization and their involvement with biological processes. Gene expression data of rice *CAMTAs* disclosed the significant involvement of *OsCAMTAs* to various biotic and abiotic vagaries. The findings are important for breeding and stress-biology program for one of the most economical crops of the globe to feed the billion.

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Author contribution HG, AD, and JB conceptualized the idea of these research analysis and article. HG has done the required analysis and prepared the draft manuscript. AD, SBD, and JB reviewed and edited the manuscript. DN and DK helped in making some of the analysis. All authors contributed to the article and approved the submitted version.

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Declarations

Conflict of interest The authors declare no competing interests.

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